



Chronic Obstructive Pulmonary Disease (COPD)

(Chronic Obstructive Bronchitis; Emphysema)

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Chronic obstructive pulmonary disease (COPD) is airflow limitation caused by an inflammatory response to inhaled toxins, often cigarette smoke. Alpha-1 antitrypsin deficiency and various occupational exposures are less common causes in patients who do not smoke. Symptoms are productive cough and dyspnea that develop over years; common signs include decreased breath sounds, prolonged expiratory phase of respiration, and wheezing. Severe cases may be complicated by weight loss, pneumothorax, frequent acute decompensation episodes, right heart failure, and/or acute or chronic respiratory failure. Diagnosis is based on history, physical examination, chest radiograph, and pulmonary function tests. Treatment is with bronchodilators, corticosteroids, and, when necessary, oxygen and antibiotics. Lung volume reduction procedures or transplantation are used in advanced disease. Survival in COPD is related to the severity of airflow limitation, the frequency of exacerbations, and the presence of comorbidities.

COPD comprises

- Chronic obstructive bronchitis (clinically defined)
- Emphysema (pathologically or radiologically defined)

Many patients have features of both.

Chronic obstructive bronchitis is chronic bronchitis with airflow obstruction. Chronic bronchitis is defined as productive cough on most days of the week for at least 3 months total duration in 2 successive years. Chronic bronchitis becomes chronic obstructive bronchitis if spirometric evidence of airflow obstruction develops. Chronic asthmatic bronchitis is a similar, overlapping condition characterized by chronic productive cough, wheezing, and partially reversible airflow obstruction; it occurs predominantly in patients who smoke and have a history of [asthma](#). When asthma and COPD co-exist in the same patient, pharmacologic treatment should follow guidelines for asthma ([1](#)).

Emphysema is destruction of lung parenchyma leading to loss of elastic recoil and loss of alveolar septa and radial airway traction, which increases the tendency for airway collapse. Lung hyperinflation, airflow limitation, and air trapping follow. Airspaces enlarge and may eventually develop blebs or bullae. Obliteration of small airways is thought to be the earliest lesion that precedes the development of emphysema.

General reference

1. [Global Initiative for Chronic Obstructive Lung Disease \(GOLD\)](#): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2024 report.

Epidemiology of COPD

In the United States, approximately 24 million people have airflow limitation, of whom approximately 16 million have a diagnosis of COPD ([1](#)). COPD is a leading cause of death, resulting in approximately 140,000 deaths each year in the United States ([2](#)). Prevalence, incidence, and mortality rates increase with age. Prevalence is higher in females, but total mortality is similar in both sexes. COPD seems to aggregate in families independent of [alpha-1 antitrypsin deficiency](#) (alpha-1 antiprotease inhibitor deficiency).

COPD is increasing worldwide because of increases in smoking and reduction in mortality due to infectious diseases. In some regions, the widespread use of biomass fuels, such as wood, grasses, or other organic materials, also contributes to COPD prevalence. COPD mortality rates may be higher in medically underserved nations than in nations where medical care is more easily accessed. COPD accounted for 3.23 million deaths globally in 2019 and is the third leading cause of death.

The COVID-19 pandemic posed a particular risk to patients with COPD. The mortality rate for patients with COPD and COVID-19 was 15% versus 4% in those without COPD ([3](#)).

Epidemiology references

1. [Centers for Disease Control and Prevention](#): Chronic Obstructive Pulmonary Disease (COPD). Updated June 30, 2023.
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3. [Meza D, Khuder B, Bailey JJ, et al](#): Mortality from COVID-19 in patients with COPD: A US study in the N3C Data Enclave. *Int J Chron Obstruct Pulmon Dis* 16:2323–2326, 2021. doi: 10.2147/COPD.S318000

Etiology of COPD

There are 2 main causes of COPD:

- Smoking (and less often other inhalational exposures)
- Genetic factors

Smoking and other inhalational exposures

Of all inhalational exposures, cigarette smoking is the primary risk factor in most countries, although only approximately 15% of people who smoke develop clinically apparent COPD (1). The risk for COPD increases with both duration (years of smoking) and cumulative dose (pack-years) (2). People who smoke and have preexisting airway reactivity (defined by increased sensitivity to inhaled methacholine), even in the absence of clinical [asthma](#), are at greater risk of developing COPD than are those without.

Smoke from indoor cooking and heating is an important causative factor in countries where indoor fires are commonly used for cooking or heating (3).

Exposure to passive cigarette smoke, [air pollution](#), and [occupational dust](#) (eg, mineral dust, cotton dust) or inhaled chemicals (eg, cadmium) contribute to the risk of COPD but are of less importance compared to cigarette smoking. Low body weight and childhood respiratory disorders also contribute to the risk of COPD.

Genetic factors

The best-defined causative genetic disorder is [alpha-1 antitrypsin deficiency](#), which is an important cause of emphysema in people who do not smoke and markedly increases susceptibility to disease in people who do.

More than 30 genetic alleles have been found to be associated with COPD or decline in lung function in selected populations, but none has been shown to be as consequential as alpha-1 antitrypsin.

Etiology references

1. [Wheaton AG, Liu Y, Croft JB, et al](#): Chronic Obstructive Pulmonary Disease and Smoking Status - United States, 2017. *MMWR Morb Mortal Wkly Rep* 68(24):533–538, 2019. doi:10.15585/mmwr.mm6824a1
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3. [Ortiz-Quintero B, Martínez-Espinosa J, Pérez-Padilla R](#): Mechanisms of Lung Damage and Development of COPD Due to Household Biomass-Smoke Exposure: Inflammation, Oxidative Stress, MicroRNAs, and Gene Polymorphisms. *Cells* 12(1):67, 2022. doi:10.3390/cells12010067

Pathophysiology of COPD

Various factors cause the airflow limitation and other complications of COPD.

Inflammation

Inhalational exposures can trigger an inflammatory response in airways and alveoli that leads to disease in genetically susceptible people. The process is thought to be mediated by an increase in protease activity and a decrease in antiprotease activity. Lung proteases, such as neutrophil elastase, matrix metalloproteinases, and cathepsins, break down elastin and connective tissue in the normal process of tissue repair. Their activity is normally balanced by antiproteases, such as alpha-1 antitrypsin, airway epithelium-derived secretory leukoprotease inhibitor, elafin, and matrix metalloproteinase tissue inhibitor. In patients with COPD, activated neutrophils and other inflammatory cells release proteases as part of the inflammatory process; protease activity exceeds antiprotease activity, and tissue destruction and mucus hypersecretion result.

Activation of neutrophils and macrophages also leads to accumulation of free radicals, superoxide anions, and hydrogen peroxide, which inhibit antiproteases and cause bronchoconstriction, mucosal edema, and mucous hypersecretion. Neutrophil-induced oxidative damage, release of profibrotic neuropeptides (eg, bombesin), and reduced levels of vascular endothelial growth factor (VEGF) may contribute to apoptotic destruction of lung parenchyma.

The inflammation in COPD increases as disease severity increases, and, in severe (advanced) disease, inflammation does not resolve completely despite smoking cessation. This chronic inflammation does not seem to respond to corticosteroids, particularly in patients who continue to smoke cigarettes (1).

Mechanical stress on alveoli from over-distension may make them susceptible to proteases, leading to alveolar septal destruction and progressive emphysema. This is particularly notable in the upper lobes of the lung (2).

Infection

Respiratory infection (to which patients with COPD are prone) may amplify progression of lung destruction.

Bacteria, especially [Haemophilus influenzae](#), colonize the lower airways of approximately 30% of patients with COPD (3). In more severely affected patients (eg, those with previous hospitalizations), colonization with [Pseudomonas aeruginosa](#) or other gram-negative bacteria is common. Smoking and airflow obstruction may lead to impaired mucus clearance in lower airways, which predisposes to infection. Repeated bouts of infection increase the inflammatory burden that hastens disease progression. There is no evidence, however, that long-term use of antibiotics slows the progression of COPD.

Airflow limitation

The cardinal pathophysiologic feature of COPD is airflow limitation caused by airway narrowing and/or obstruction, loss of elastic recoil, or both.

Airway narrowing and obstruction are caused by inflammation-mediated mucus hypersecretion, mucus plugging, mucosal edema, bronchospasm, peribronchial fibrosis, and remodelling of small airways or a combination of these mechanisms. Alveolar septa are destroyed, reducing parenchymal attachments to the airways and thereby facilitating airway closure during expiration.

Enlarged alveolar spaces sometimes consolidate into bullae, defined as airspaces ≥ 1 cm in diameter. Bullae may be entirely empty or have strands of lung tissue traversing them in areas of locally severe emphysema; they occasionally occupy an entire hemithorax. These changes lead to loss of elastic recoil and lung hyperinflation.

Increased airway resistance increases the work of breathing. Lung hyperinflation, although it decreases airway resistance, also increases the work of breathing. Increased work of breathing may lead to alveolar hypoventilation with hypoxia and hypercapnia, although hypoxia and hypercarbia can also be caused by ventilation/perfusion (V/Q) mismatch.

Complications

In addition to airflow limitation and sometimes respiratory insufficiency, complications include

- Pulmonary hypertension
- Respiratory infection
- Weight loss and other comorbidities

Chronic alveolar hypoxia increases pulmonary vascular tone, which, if diffuse, causes [pulmonary hypertension](#) and [cor pulmonale](#). The increase in pulmonary vascular pressure may be augmented by the destruction of the pulmonary capillary bed due to destruction of alveolar septa.

Viral or bacterial respiratory infections are common among patients with COPD and cause a large percentage of acute exacerbations. It is currently thought that acute bacterial infections are due to acquisition of new strains of bacteria rather than overgrowth of chronic colonizing bacteria.

Weight loss may occur, perhaps in response to insufficient caloric intake and increased levels of circulating tumor necrosis factor (TNF)-alpha. This weight loss may be due to a mismatch between caloric expenditure and nutritional intake because caloric expenditure can be high in the presence of heightened inflammatory cytokines and hypoxemia.

Other coexisting or complicating disorders that adversely affect quality of life and/or survival include [osteoporosis](#), [depression](#), [anxiety](#), [coronary artery disease](#), [arrhythmias](#), [lung cancer](#) and other cancers, muscle atrophy, and [gastroesophageal reflux](#). The extent to which these disorders are consequences of COPD, smoking, and the accompanying systemic inflammation is unclear.

Pathophysiology references

1. [Adcock IM, Bhatt SP, Balkissoon R, Wise RA](#): The Use of Inhaled Corticosteroids for Patients with COPD Who Continue to Smoke Cigarettes: An Evaluation of Current Practice. *Am J Med* 135(3):302–312, 2022. doi:10.1016/j.amjmed.2021.09.006
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Symptoms and Signs of COPD

COPD takes years to develop and progress.

- Productive cough usually is the initial symptom, developing among people who smoke and are in their 40s and 50s.
- [Dyspnea](#) that is progressive, persistent, exertional, or worse during respiratory infection appears when patients are in their late 50s or 60s.

Symptoms usually progress quickly in patients who continue to smoke and in those who have a higher lifetime tobacco exposure. Morning headache develops in more advanced disease and signals nocturnal hypercapnia or hypoxemia.

Signs of COPD include wheezing, a prolonged expiratory phase of breathing, lung hyperinflation manifested as decreased heart and lung sounds, and increased anteroposterior diameter of the thorax (barrel chest).

Signs of advanced disease include pursed-lip breathing, accessory respiratory muscle use, paradoxical inward movement of the lower rib cage during inspiration (Hoover sign), and cyanosis. Signs of cor pulmonale include neck vein distention, splitting of the second heart sound with an accentuated pulmonic component, tricuspid insufficiency murmur, and peripheral edema. Right ventricular heaves are uncommon in COPD because the lungs are hyperinflated.

Patients with advanced emphysema lose weight and experience muscle wasting that has been attributed to immobility, hypoxia, or release of systemic inflammatory mediators, such as TNF-alpha.

Spontaneous pneumothorax may occur (possibly related to rupture of bullae) and should be suspected in any patient with COPD whose pulmonary status abruptly worsens.

The symptoms can be graded according to which activities cause dyspnea (see table [Breathlessness Measurement using the mMRC Questionnaire](#)).

TABLE

Breathlessness Measurement Using the Modified British Medical Research Council (mMRC) Questionnaire

Grade	Shortness of Breath
0	None except during strenuous exercise
1	Occurring when hurrying on level ground or walking up a slight incline
2	Resulting in walking more slowly than people of the same age on level ground <i>or</i> Resulting in stopping for breath when walking at own pace on level ground
3	Resulting in stopping for breath after walking about 100 meters or after a few minutes on level ground
4	Preventing the person from leaving the house <i>or</i> Occurring when dressing or undressing

Adapted from Mahler DA, Wells CK: Evaluation of clinical methods for rating dyspnea. *Chest* 93:580–586, 1988.

Acute exacerbations

Acute exacerbations occur sporadically during the course of COPD and are heralded by increased symptom severity. The specific cause of any exacerbation is almost always impossible to determine, but exacerbations are often attributed to viral upper respiratory infections, acute bacterial bronchitis, or exposure to respiratory irritants. As COPD progresses, acute exacerbations tend to become more frequent, averaging approximately 1 to 3 episodes/year.

Diagnosis of COPD

- Chest imaging
- Pulmonary function testing

Diagnosis is suggested by history, physical examination, and chest imaging findings and is confirmed by [pulmonary function tests](#). Similar symptoms can be caused by [asthma](#), [heart failure](#), and [bronchiectasis](#) (see table [Differential Diagnosis of COPD](#)). COPD and asthma are sometimes easily confused and may coexist.

TABLE

Differential Diagnosis of COPD

Diagnosis	Onset	Imaging Results	Other Features
COPD	Middle age	Sometimes lung hyperinflation, bullae, increased retrosternal air space, and/or bronchial wall thickening (seen on chest imaging); however, imaging usually is not helpful diagnostically and is done mainly to exclude other disorders	Slowly progressive symptoms History of smoking or exposure to tobacco or other types of smoke
Asthma	Early in life (often during childhood)	Usually normal or possibly hyperinflation	Symptoms vary widely from day to day Variable airflow obstruction Symptoms often worse at night or early morning History of allergies, rhinitis, or eczema Often family history of asthma

Diagnosis	Onset	Imaging Results	Other Features
Bronchiectasis	All ages, but most often in older or middle aged adults	Bronchial dilation and bronchial wall thickening (seen on chest radiograph or chest CT)	Often large amounts of purulent sputum Often history of chronic recurrent bacterial infection
Diffuse panbronchiolitis	Usually between ages 10 and 60 years (mean age of 40)	Diffuse small centrilobular nodular opacities and hyperinflation seen on chest radiograph and high-resolution CT	Mostly males who do not smoke Almost all have chronic sinusitis Predominately in those of Asian descent
Heart failure	All ages, but most often in older or middle age	Enlarged heart, pleural effusion, fluid in major fissure, sometimes pulmonary edema (seen on chest radiograph)	Volume restriction without airflow limitation (detected by pulmonary function tests)
Obliterative bronchiolitis	Onset at younger age	Peripheral hypodense areas (seen on chest CT during expiration)	History of lung or bone marrow transplantation
Tuberculosis (TB)	All ages	Lung infiltrates seen on chest radiograph	Confirmed by microbiologic testing Usually in areas with high prevalence of TB
Data adapted from Global Initiative for Chronic Obstructive Lung Disease (GOLD) : Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2024 report.			

Systemic disorders that may have a component of airflow limitation suggest COPD; they include [HIV infection](#), abuse of illicit drugs (particularly cocaine and amphetamines), [sarcoidosis](#), [Sjögren syndrome](#), [bronchiolitis obliterans](#), [lymphangioleiomyomatosis](#), and eosinophilic granuloma. COPD can be differentiated from [interstitial lung diseases](#) by chest imaging, which shows increased interstitial markings in interstitial lung disease, and pulmonary function testing, which shows a restrictive ventilatory defect rather than an obstructive ventilatory defect. In some patients, COPD and interstitial lung disease coexist (combined pulmonary fibrosis and emphysema [CPFE]) in which lung volumes are relatively preserved, but gas exchange is severely impaired.

Pulmonary function tests

Patients suspected of having COPD should undergo [pulmonary function testing](#) to confirm airflow limitation, to quantify its severity and reversibility, and to distinguish COPD from other disorders. (Some experts recommend screening pulmonary function testing for all patients with a history of smoking.) Pulmonary function testing is also useful for following disease progression and monitoring response to treatment. The primary diagnostic tests are

- FEV1: The volume of air forcefully expired during the first second after taking a full breath
- Forced vital capacity (FVC): The total volume of air expired with maximal force
- Flow-volume loops: Simultaneous spirometric recordings of airflow and volume during forced maximal expiration and inspiration

Reductions of FEV1, FVC, and the ratio of FEV1/FVC are the hallmark of airflow limitation. Flow-volume loops show a concave pattern in the expiratory tracing.

There are 2 basic pathways by which COPD can develop and manifest with symptoms in later life:

- In the first pathway, patients may have normal lung function in early adulthood, which is followed by a more rapid decline in FEV1 (≥ 60 mL/year).
- With the second pathway, patients have impaired lung function in early adulthood, often associated with asthma or other childhood respiratory disease. In these patients, COPD may present with a normal age-related decline in FEV1 (approximately 30 mL/year).

Although this second pathway model is conceptually helpful, a wide range of individual trajectories is possible (1). When the FEV1 falls below approximately 1 L, patients develop dyspnea during activities of daily living (although dyspnea is more closely related to the degree of dynamic hyperinflation [progressive hyperinflation due to incomplete exhalation] than to the degree of airflow limitation). When the FEV1 falls below approximately 0.8 L, patients are at risk of hypoxemia, hypercapnia, and [cor pulmonale](#).

FEV1 and FVC are easily measured with office spirometry. Normal reference values are determined by patient age, sex, and height. It is recommended that race not be used for calculating predicted reference values (2). Airflow limitation severity in patients with COPD and FEV1/FVC < 0.70 can be classified based on post-bronchodilator FEV1 (3):

- Mild: $\geq 80\%$ of predicted
- Moderate: 50% to 79% of predicted
- Severe: 30% to 49% of predicted
- Very severe: < 30% of predicted

Additional pulmonary function testing is necessary only in specific circumstances, such as before [lung volume reduction procedures](#). Other test abnormalities may include

- Increased total lung capacity
- Increased functional residual capacity
- Increased residual volume
- Decreased vital capacity
- Decreased single-breath diffusing capacity for carbon monoxide (DLCO)

Findings of increased total lung capacity, functional residual capacity, and residual volume can help distinguish COPD from restrictive pulmonary disease, in which these measures are diminished.

Decreased DLCO is nonspecific and is reduced in other disorders that affect the pulmonary vascular bed, such as interstitial lung disease, but can help distinguish emphysema from asthma, in which DLCO is normal or elevated.

Imaging tests

Chest radiograph may have characteristic findings. In patients with emphysema, changes can include lung hyperinflation manifested as a flat diaphragm (ie, increase in the angle formed by the sternum and anterior diaphragm on a lateral film from the normal value of 45° to > 90°), rapid tapering of hilar vessels, and bullae (ie, radiolucencies > 1 cm surrounded by arcuate, hairline shadows). Other typical findings include enlargement of the retrosternal airspace and a narrow cardiac shadow. Emphysematous changes occurring predominantly in the lung bases suggest [alpha-1 antitrypsin deficiency](#). The lungs may look normal or have increased lucency secondary to loss of parenchyma. Among patients with chronic obstructive bronchitis, chest radiographs may be normal or may show a bibasilar increase in bronchovascular markings as a result of bronchial wall thickening.

Chronic Obstructive Pulmonary Disease (Chest Radiograph)

IMAGE

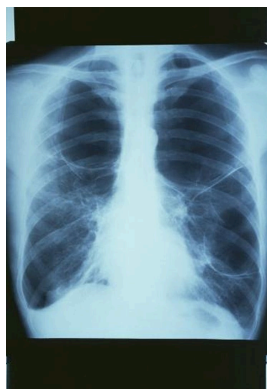


Chest radiograph of a patient with chronic obstructive pulmonary disease (COPD). The lungs are hyperinflated, the diaphragm is flattened, vascular markings are increased, and the heart size is marginally increased.

BY PERMISSION OF THE PUBLISHER. FROM BARNES P. IN *BONE'S ATLAS OF PULMONARY AND CRITICAL CARE MEDICINE*. EDITED BY J CRAPO. PHILADELPHIA, CURRENT MEDICINE, 2005.

COPD with Bullae

IMAGE



This chest radiograph shows a large bulla in the upper right lung and 2 large bullae in the left lung.

GJLP/CNRI/SCIENCE PHOTO LIBRARY

Prominent hila suggest large central pulmonary arteries that may signify [pulmonary hypertension](#). Right ventricular enlargement that occurs in [cor pulmonale](#) may be masked by lung hyperinflation or may manifest as encroachment of the heart shadow on the retrosternal space or by widening of the transverse

cardiac shadow in comparison with previous chest radiographs.

Chest CT may reveal abnormalities that are not apparent on the chest radiograph and may also suggest coexisting or complicating disorders, such as [pneumonia](#), [pneumoconiosis](#), or [lung cancer](#). CT helps assess the extent and distribution of emphysema, estimated either by visual scoring or with analysis of the distribution of lung density. Indications for obtaining CT in patients with COPD include evaluation for lung volume reduction procedures, suspicion of coexisting or complicating disorders that are not clearly evident or excluded by chest radiograph, suspicion of lung cancer, and [screening for lung cancer](#). Enlargement of the pulmonary artery diameter greater than the ascending aorta diameter suggests pulmonary hypertension (4).

Adjunctive tests

Alpha-1 antitrypsin levels should be measured in patients < 50 years with symptomatic COPD and in patients of any age with COPD who do not smoke to detect [alpha-1 antitrypsin deficiency](#). Other indications of possible alpha-1 antitrypsin deficiency include a family history of premature COPD or unexplained liver disease, lower-lobe distribution of emphysema, and COPD associated with antineutrophil cytoplasmic antibody (ANCA)-positive vasculitis. If levels of alpha-1 antitrypsin are low, the diagnosis should be confirmed by genetic testing to establish the alpha-1 antitrypsin phenotype.

ECG, often done to exclude cardiac causes of dyspnea, typically shows diffusely low QRS voltage with a vertical heart axis caused by lung hyperinflation and increased P-wave voltage or rightward shifts of the P-wave vector caused by right atrial enlargement in patients with advanced emphysema. Findings of right ventricular hypertrophy include an R or R' wave as tall as or taller than the S wave in lead V1; an R wave smaller than the S wave in lead V6; right-axis deviation > 110° without right bundle branch block; or some combination of these findings. Multifocal [atrial tachycardia](#), an arrhythmia that can accompany COPD, manifests as a tachyarrhythmia with polymorphic P waves and variable PR intervals.

Echocardiography is occasionally useful for assessing right ventricular function and pulmonary hypertension, although air trapping makes it technically difficult in patients with COPD. Echocardiography is most often indicated when coexistent left ventricular or valvular heart disease is suspected.

Hemoglobin and hematocrit are of little diagnostic value in the evaluation of COPD but may show erythrocythemia (hematocrit > 48%) if the patient has chronic hypoxemia. Patients with anemia (for reasons other than COPD) have disproportionately severe dyspnea. The differential white blood cell count may be helpful. A growing body of evidence indicates that [eosinophilia](#) predicts response to inhaled corticosteroids.

Serum electrolytes are of little value but may show an elevated bicarbonate level if patients have chronic hypercapnia. Venous blood gases are useful for diagnosis of acute or chronic hypercapnia.

Evaluation of exacerbations

Patients with acute exacerbations usually have combinations of increased cough, sputum, dyspnea, and work of breathing, as well as low oxygen saturation determined with pulse oximetry, diaphoresis, tachycardia, anxiety, and cyanosis. Patients with exacerbations accompanied by retention of carbon dioxide may be lethargic or somnolent, a very different appearance.

All patients requiring hospitalization for an acute exacerbation should undergo testing to quantify hypoxemia and hypercapnia. Hypercapnia may exist without hypoxemia.

Findings of partial pressure of arterial oxygen (PaO₂) < 50 mm Hg, or partial pressure of carbon dioxide in arterial blood (PaCO₂) > 50 mm Hg, or partial pressure of carbon dioxide in venous blood (PvCO₂) > 55 mm Hg in patients with respiratory acidemia (pH < 7.35) define [acute respiratory failure](#). Some patients chronically manifest such levels of PaO₂ and PaCO₂ in the absence of acute respiratory failure.

A [chest radiograph](#) is often done to check for [pneumonia](#) or [pneumothorax](#). Point of care ultrasound (POCUS) may prove to be a useful adjunctive procedure for rapid diagnosis of pneumonia or pneumothorax. In patients with acute onset of symptoms, chest CT angiogram is done to check for [pulmonary embolism](#). Very rarely, among patients receiving chronic systemic corticosteroids, infiltrates may represent [Aspergillus pneumonia](#).

Yellow or green sputum is a reliable indicator of neutrophils in the sputum and suggests bacterial colonization or infection.

Sputum culture is usually done in hospitalized patients but is not usually necessary in outpatients. In samples from outpatients, Gram stain usually shows neutrophils with a mixture of organisms, often gram-positive diplococci (*Streptococcus pneumoniae*), gram-negative bacilli (*H. influenzae*), or both. However, culture and microscopic examination of sputum is usually not necessary for outpatients. Other oropharyngeal commensal organisms, such as *Moraxella catarrhalis* (formerly known as *Branhamella catarrhalis*), occasionally cause exacerbations. In hospitalized patients, resistant gram-negative organisms (eg, *Pseudomonas*) or, rarely, *Staphylococcus* may be identified in culture specimens.

During influenza season, a rapid influenza test will guide treatment with anti-influenza agents, and a respiratory viral panel for respiratory syncytial virus (RSV), rhinovirus, and metapneumovirus may allow tailoring of antimicrobial therapy. Testing for [COVID-19](#) and consideration of COVID-19-specific therapies is also indicated.

Serum C-reactive protein (CRP) is helpful in guiding the use of antibiotics during exacerbations. In clinical trials, use of antibiotics can be decreased in patients with low CRP without evidence of harm (5,6).

Diagnosis references

1. [Lange P, Celli B, Agusti A, et al](#): Lung-function trajectories leading to chronic obstructive pulmonary disease. *N Engl J Med* 373(2):111–122, 2015.
2. [Bhakta NR, Bime C, Kaminsky DA, et al](#): Race and ethnicity in pulmonary function test interpretation: An Official American Thoracic Society Statement. *Am J Respir Crit Care Med* 207(8):978–995, 2023. doi:10.1164/rccm.202302-0310ST
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6. [Prins HJ, Duijkers R, van der Valk P, et al](#): CRP-guided antibiotic treatment in acute exacerbations of COPD in hospital admissions. *Eur Respir J* 53(5):1802014, 2019. doi: 10.1183/13993003.02014-2018

Treatment of COPD

(See also [Treatment of Stable COPD](#) and [Treatment of Acute COPD Exacerbation](#).)

- Smoking cessation
- Inhaled bronchodilators, corticosteroids, or both
- Supportive care (eg, oxygen therapy, pulmonary rehabilitation)

COPD management involves treatment of [chronic stable disease](#) and prevention and treatment of [exacerbations](#). [Treatment of cor pulmonale](#), a common complication of long-standing, severe COPD, is discussed elsewhere.

[Smoking cessation](#) is critical in treatment of COPD.

Treatment of chronic stable COPD aims to prevent exacerbations and improve lung and physical function. Treatment relieves symptoms rapidly with primarily short-acting beta-adrenergic medications and decreases exacerbations with inhaled corticosteroids, long-acting beta-adrenergic medications, long-acting anticholinergic medications, or a combination (see table [Pharmacotherapy of COPD](#)).

[Pulmonary rehabilitation](#) includes structured and supervised exercise training, nutrition counseling, and self-management education.

[Oxygen therapy](#) is indicated for selected patients.

[Treatment of exacerbations](#) ensures adequate oxygenation and near-normal blood pH, reverses airway obstruction, and treats any cause.

Immunization against [influenza](#), [pneumococcus](#), [COVID-19](#), and [respiratory syncytial virus](#) (RSV—in patients age 60 years and older) should also be recommended as a preventive measure.

End-of-life care

In patients with very severe disease, exercise is unwarranted and activities of daily living are arranged to minimize energy expenditure. For example, patients may arrange to live on one floor of the house, have several small meals rather than fewer large meals, and avoid wearing shoes that must be tied. [End-of-life care](#) should be discussed, including whether to pursue mechanical ventilation, the use of palliative sedation, and appointment of a surrogate medical decision-maker in the event of the patient's incapacitation.

Prognosis for COPD

Severity of airway obstruction predicts survival in patients with COPD. For an FEV1 35 to 55% predicted, the 5-year mortality is 40% ([1](#)). For an FEV1 less than 35% predicted, the 5-year mortality is 55% ([1](#)).

More accurate prediction of risk of death is possible by simultaneously measuring body mass index (*B*), the degree of airflow obstruction (*O*, which is the FEV1), dyspnea (*D*, which is measured using the [Modified British Medical Research Council \(mMRC\) Questionnaire](#)), and exercise capacity (*E*, which is measured with a [6-minute walk test](#)); this is the BODE index. Also, older age and the presence of heart disease, anemia, resting tachycardia, hypercapnia, or hypoxemia predict decreased survival, whereas a significant response to bronchodilators predicts improved survival. Risk factors for death in patients with acute exacerbation requiring hospitalization include older age, higher PaCO₂, and use of maintenance oral corticosteroids.

CLINICAL CALCULATORS

[BODE Index for COPD Survival Prediction](#)



Patients at high risk of imminent death are those with progressive unexplained weight loss or severe functional decline (eg, those who experience dyspnea with self-care, such as dressing, bathing, or eating).

Mortality in COPD may result from comorbidities (eg, cardiovascular disease, cancer) or intercurrent illnesses rather than from progression of the underlying disorder in patients who have stopped smoking. Death is generally caused by acute respiratory failure, pneumonia, lung cancer, heart disease, or pulmonary embolism.

Prognosis reference

1. [Almagro P, Martinez-Camblor P, Soriano JB, et al](#): Finding the best thresholds of FEV1 and dyspnea to predict 5-year survival in COPD patients: the COCOMICS study. *PLoS One* 9(2):e89866, 2014. doi: 10.1371/journal.pone.0089866

Key Points

- Cigarette smoking in susceptible people is the major cause of chronic obstructive pulmonary disease (COPD) in the developed world.
- Diagnose COPD and differentiate it from disorders that have similar characteristics (eg, asthma, heart failure) primarily by routine clinical information, such as symptoms (particularly time course), age at onset, risk factors, and results of routine tests (eg, chest radiograph, pulmonary function tests).
- Reductions of FEV1, FVC, and the ratio of FEV1/FVC post-bronchodilator are characteristic findings.
- Categorize patients based on symptoms and exacerbation risk into one of 3 groups and use that category to guide treatment.
- Relieve symptoms rapidly with primarily short-acting beta-adrenergic medications and decrease exacerbations with inhaled corticosteroids, long-acting beta-adrenergic medications, long-acting anticholinergic medications, or a combination.
- Encourage smoking cessation using multiple interventions.
- Immunization against influenza, pneumococcus, COVID-19, and respiratory syncytial virus ([RSV] in patients age ≥ 60 years) should also be done as a preventive measure.



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